

Week 10 • Where the state goes: extraction and lifting

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Refactoring by Extracting Components

Source: [Lectures/Technical/Structure/Refactoring-by-Extracting-Components.md](#) — something unclear? [Open an issue](#), or send a pull request.

Lesson: refactoring is critical for having a codebase that is maintainable and pleasant to work with

Challenge: Compare the following equivalent versions of code

- They have **exactly the same behavior**: rendering items in the todo list
- Which of them is easier to read?

Version 1

```

160 <ul>
161   {list.map((elem) => (
162     <div>
163       <li
164         style={{
165           listStyle: "none",
166           backgroundColor:
167             elem.id === activeElementId
168               ? "lightgoldenrodyellow"
169               : "white",
170           fontSize: elem.id === activeElementId ? "x-large" : "1em",
171           color: elem.id === activeElementId ? "violet" : "darkBlue",
172           fontWeight: elem.id === activeElementId ? "700" : "400",
173         }}
174         key={elem.id}
175       >
176         {!elem.done ? (
177           elem.id !== activeElementId ? (
178             <PlayArrowOutlined
179               onClick={() => handleStartTimer(elem.id)}
180             ></PlayArrowOutlined>
181           ) : (
182             <StopCircleOutlined
183               onClick={() => handleStopTimer(elem.id)}
184             ></StopCircleOutlined>
185           )
186         ) : (
187           <PlayArrowOutlined
188             style={{
189               visibility: "hidden",
190             }}
191           ></PlayArrowOutlined>
192         )}
193
194         {elem.name}
195
196         { /* Checkbox */ }
197         <input
198           type="checkbox"
199           value={elem.done}
200           checked={elem.done}
201           onChange={() => handleCheckbox(elem.id)}
202         />
203         {elem.done && " 🏁"}

```

```

204
205     {(() => {
206       const displayTime = elem.currentSessionStart
207       ? (elem.totalTime || 0) +
208         Math.floor((Date.now() - elem.currentSessionStart) / 1000)
209       : elem.totalTime || 0;
210       return displayTime > 0 ? (
211         <span
212           style={{
213             fontSize: "small",
214             color: "green",
215             marginLeft: "1em",
216           }}
217         >
218           {moment.duration(displayTime, "seconds").humanize()}{" "}
219           {elem.id === activeElementId && (
220             <AccessTime
221               style={{
222                 fontSize: "14px",
223                 transform: `rotate(${(displayTime % 4) * 90}deg)`,
224                 transition: "transform 0.3s ease",
225               }}
226             />
227           )}
228         </span>
229       ) : null;
230     })()}
231     { /* Remove */ }
232     <a
233       href="#"
234       onClick={() => deleteElement(elem.id)}
235       style={{
236         color: "■ #a2a2a2cf",
237         fontSize: "xx-small",
238         marginLeft: "2em",
239         textDecoration: "none",
240       }}
241     >
242       (remove)
243     </a>
244   </li>
245 </div>
246 ))}
247 </ul>

```

Version 2

```
170 <ul>
171   {list.map((elem) => (
172     <ToDoItem highlight={elem.id == activeElementId} key={elem.id}>
173       <PlayStopButton
174         isHidden={elem.done}
175         isPlaying={elem.id === activeElementId}
176         onPlay={() => handleStartTimer(elem.id)}
177         onStop={() => handleStopTimer(elem.id)}
178       />
179
180       {elem.name}
181
182       <input
183         type="checkbox"
184         checked={elem.done}
185         onChange={() => handleCheckbox(elem.id)}
186       />
187
188       {elem.done && " 🏁"}
189
190       <ItemDuration
191         displayTime={totalTime(elem)}
192         showAnimation={elem.id === activeElementId}
193       />
194
195       <RemoveItemButton onClick={() => deleteElement(elementId)} />
196     </ToDoItem>
197   )]}
198 </ul>
```

- this version allows us to think in terms of the UI elements that are all at the same abstraction level - the play/stop button, the name of the todo list item, its duration, the remove button.
- the previous version was at all possible abstraction levels
 - play button
 - maths computation
 - css details
 - etc.

Principle: Code in a component should be at the same abstraction level

Advice: iterate. iterate. iterate

- there is no way to arrive to simple code without a lot of iteration

Advice 2: first make it work. then refactor it and make it nice

- you never get it right from the first time

Where the State Goes

Extraction has a consequence. Split one component into three and some of them will need to agree about the same piece of state — which now belongs to none of them. The two patterns from [Component Communication](#) (props down, callbacks up) are not enough on their own.

sibling to sibling by “*lifting the state*”

Sometimes, you want the state of two components to always change together.

To do it, you *lift the state up*:

- remove state from both of them
- move it to their closest common parent, and then
- pass it down to them via props

This is a common thing you will do when writing React

Example: in the TODO list application, ensuring that one can’t start two tasks in two different lists

Sending a message from one component to the other

- also has to be done via the parent
- in our case, imagine that you want to move one of the tasks, from one list to the other
- the parent would have to know about all the lists and a given list will have to call a callback on the parent to decide to which list does one

To read for next time: [Sharing state between components](#)

- talks about lifting state with an example

Exam Questions

1. What does “lifting state up” mean in React?

2. Why is this principle important: “*Code in a component should be at the same abstraction level*”?

Something unclear in this section? [Open an issue](#).

The same function, written twice

Components are not the only thing worth extracting. A common case: `const getCurrentUser = async () => ...` defined once in `ProfileBar` and again in `Settings`, because both screens needed it and neither knew about the other.

Two copies of a function are two things to fix when the backend changes, and they drift silently — one gets a null check, the other does not. Pull it into a **custom hook** (`useCurrentUser`) and both screens call the same code.

The signal to watch for: you are about to write something you have a feeling you already wrote. You probably did.

Something unclear in this section? [Open an issue](#).

The `useRef` hook

Source: [Lectures/Technical/React/The-useRef-Hook.md](#) — something unclear? [Open an issue](#), or send a pull request.

Allows you to get hold of a DOM element and manipulate it

Example: Automatically focusing a given element on first render

```
import {useRef, useEffect} from 'react'

const InputRefExample = () => {

  const inputRef = useRef(null);

  useEffect(() => {
    inputRef.current?.focus()
  }, [])

  return(
    <div>
      <input ref={inputRef} />
      <button>
        Submit
      </button>
    </div>
  )
}

export default InputRefExample
```

Canonical use of the `useRef` hook

1. Import `useRef`
2. Declare a ref object (i.e. `inputRef`) - initialize it with null, it's fine
3. Pass your object as the `ref` attribute to the desired DOM element
4. Refer to the DOM object by using the `current` property

Other Examples of manipulating the DOM

- scrolling to a given node
- getting the dimensions of a given DOM node
- keeping track of a timer (see the TODO list application)

Notes

Optional chaining to prevent errors

The optional chaining (`?.`) prevents errors if the element isn't mounted yet.

```
useEffect(() => {  
  inputRef.current?.focus()  
}, [])
```

The reference is persisted between renders

State changes of the referenced element are not observed by React

Read More

- [useRef in the react documentation](#)
- Nice example of [controlling a video player component](#)
 - you need to interact with the video element in the DOM
 - the DOM has already a powerful implementation of
 - get a reference to the relevant DOM element
 - call the `start` and `stop` methods

Exam Questions

1. What is the purpose of the useRef hook?
2. What does this code do and why use optional chaining?

```
const inputRef = useRef(null);  
  
useEffect(() => {  
  inputRef.current?.focus();  
}, []);  
  
return <input ref={inputRef} />;
```

Something unclear in this section? Open an issue.

Component Extraction & Semantic React Architecture

Source: [Lectures/Technical/Structure/Component-Extraction-Guide.md](#) — something unclear? [Open an issue](#), or send a pull request.

On the examples below. They are written in TypeScript with Tailwind class names, because they came from a different course. This course is JavaScript and MUI. Read them for the *decisions* — when a block deserves a name, when repetition earns a component — and ignore the syntax around them; nothing here depends on the stack.

Universal Principles (CSS-Agnostic)

These principles apply whether you use Tailwind, CSS Modules, styled-components, or plain CSS.

The Problem: “Div Soup” & “Class Soup”

Bad Example (Tailwind)

```
<div className="mx-auto flex w-full max-w-xl flex-col items-start gap-6 px-4 py-6">
  <section className="flex w-full flex-col items-center gap-2 p-6">
    <div className="size-24 rounded-3xl bg-neutral-300" />
    <h2 className="text-2xl font-bold">{user.name}</h2>
  </section>
</div>
```

Bad Example (CSS Modules)

```
<div className={styles.outerContainer}>
  <section className={styles.innerSection}>
    <div className={styles.avatar} />
    <h2 className={styles.userName}>{user.name}</h2>
  </section>
</div>
```

Bad Example (Inline Styles)

```
<div style={{ margin: '0 auto', display: 'flex', width: '100%', maxWidth: '36rem',
flexDirection: 'column', gap: '1.5rem', padding: '1rem 1rem 1.5rem 1rem' }}>
  <section style={{ display: 'flex', width: '100%', flexDirection: 'column', alignItems:
'center', gap: '0.5rem', padding: '1.5rem' }}>
    <div style={{ width: '6rem', height: '6rem', borderRadius: '1.5rem', background: '#d4d4d4'
}} />
    <h2 style={{ fontSize: '1.5rem', fontWeight: 'bold' }}>{user.name}</h2>
  </section>
</div>
```

What's the problem in ALL these examples?

- Can't tell what these divs/sections represent
 - Styling details overwhelm structure
 - Not reusable
 - Hard to review without CSS knowledge
-

The Solution: Semantic Component Extraction

Good Example (Works with ANY CSS approach)

```
<ProfileContainer>
  <ProfileHeader>
    <Avatar user={user} />
    <UserName>{user.name}</UserName>
  </ProfileHeader>
</ProfileContainer>
```

Benefits:

- Clear component hierarchy
 - Communicates intent immediately
 - Backend devs can understand structure
 - Reusable across the app
 - Styling details hidden in component files
-

The Four Core Principles

1. The Tech Lead Test

"If someone who doesn't know your CSS framework can't understand your component hierarchy, you need better abstraction."

Ask yourself: Could a backend developer review this code and understand the page structure?

2. The Semantic Naming Test

"If you can describe a component's purpose in 1-2 words, that should be its name."

```
// Bad: What is this?
<div className="...">

// Good: Immediately clear
<ProfileCard>
<NavigationMenu>
<PricingTable>
```

3. The DRY Test (Don't Repeat Yourself)

"If you copy-paste the same styling pattern more than twice, extract a component."

4. The Single Level of Abstraction (SLA)

"Keep all JSX at the same conceptual level. Don't mix high-level components with low-level HTML details."

Bad (mixed abstraction levels):

```
<ProfilePage>
  <Header />  {/* High-level component */}
  <div className="flex gap-4">  {/* Low-level HTML */}
    <div className="w-1/3 bg-gray-100 p-4">  {/* Low-level HTML */}
      <p className="font-bold">Name</p>
      <p>{user.name}</p>
    </div>
  </div>
  <Footer />  {/* High-level component */}
</ProfilePage>
```

Good (consistent abstraction level):

```
<ProfilePage>
  <Header />
  <ProfileContent>
    <ProfileSection title="Name" value={user.name} />
    <ProfileSection title="Email" value={user.email} />
  </ProfileContent>
  <Footer />
</ProfilePage>
```

Step-by-Step Tutorial: Component Extraction

Example: Refactoring a Profile Page

Step 1: Identify the Problem

Original Code (275 lines, Profile.tsx):

```
const Profile = () => {
  return (
    <div className="mx-auto flex w-full max-w-xl flex-col items-start gap-6 px-4 py-6">
      <section className="flex w-full flex-col items-center gap-2 p-6">
        <div className="size-24 rounded-3xl bg-neutral-300">
          <img src={user.avatarUrl} alt={user.name} />
        </div>
        <h2 className="text-2xl font-bold">{user.name}</h2>
        <div className="flex items-center">
          {[...Array(5)].map((_, i) => (
            <Star key={i} className="h-5 w-5 text-yellow-400" />
          ))}
        </div>
        <div className="flex items-center gap-4 text-sm text-gray-600">
          <span className="flex items-center gap-1.5">
            <Briefcase className="h-4 w-4" /> {user.role}
          </span>
          <span>• Joined {formatDate(user.joinedDate)}</span>
        </div>
        <div className="flex gap-2">
          <button className="rounded-full bg-neutral-200 px-4 py-2">Email</button>
          <button className="rounded-full bg-neutral-200 px-4 py-2">Message</button>
          <button className="rounded-full bg-neutral-200 px-4 py-2">Call</button>
        </div>
      </section>
      {/* ... 200+ more lines ... */}
    </div>
  );
};
```

Problems:

1. Everything in one file (275 lines)
 2. No semantic component names
 3. Can't understand structure without reading every line
 4. Not reusable
 5. Hard to test
-

Step 2: Identify Repeated Patterns

Ask yourself:

- What sections repeat?
- What could be reused elsewhere?
- What are the conceptual "things" on this page?

Pattern identification:

```
Profile Page
├── Container (layout wrapper)
├── Header
│   ├── Avatar
│   ├── Name
│   ├── Rating
│   ├── Metadata (role, joined date)
│   └── Action Buttons
├── About Section
├── Experience Section
│   └── Experience Entry (repeated)
├── Qualifications Section
├── Skills Section
└── Feedback Section
```

Step 3: Extract Layout Components

Create: `components/layout/ProfileContainer.tsx`

```
import { ReactNode } from 'react';

interface ProfileContainerProps {
  children: ReactNode;
}

export const ProfileContainer = ({ children }: ProfileContainerProps) => {
  return (
    <div className="mx-auto flex w-full max-w-xl flex-col items-start gap-6 px-4 py-6">
      {children}
    </div>
  );
};
```

OR with CSS Modules:

```
// ProfileContainer.module.css
.container {
  margin: 0 auto;
  display: flex;
  width: 100%;
  max-width: 36rem;
  flex-direction: column;
  gap: 1.5rem;
  padding: 1rem;
}

// ProfileContainer.tsx
import styles from './ProfileContainer.module.css';

export const ProfileContainer = ({ children }) => {
  return <div className={styles.container}>{children}</div>;
};
```

Step 4: Extract Domain Components

Create: `components/profile/ProfileHeader.tsx`

```
import { Avatar } from '@components/ui/Avatar';
import { StarRating } from '@components/ui/StarRating';
import { ProfileActions } from './ProfileActions';
import type { UserProfile } from '@types/user';

interface ProfileHeaderProps {
  user: UserProfile;
}

export const ProfileHeader = ({ user }: ProfileHeaderProps) => {
  return (
    <header className="flex w-full flex-col items-center gap-2 p-6">
      <Avatar src={user.avatarUrl} alt={user.name} size="large" />
      <h1 className="text-2xl font-bold">{user.name}</h1>
      <StarRating rating={user.rating} />
      <ProfileMetadata role={user.role} joinedDate={user.joinedDate} />
      <ProfileActions user={user} />
    </header>
  );
};
```

Benefits:

- Clear semantic name (`ProfileHeader`)
 - All header logic in one place
 - Reusable for similar profiles
 - Easy to test in isolation
-

Step 5: Extract Reusable UI Components

Create: `components/ui/Avatar.tsx`

```
interface AvatarProps {
  src: string;
  alt: string;
  size?: 'small' | 'medium' | 'large';
}

export const Avatar = ({ src, alt, size = 'medium' }: AvatarProps) => {
  const sizeClasses = {
    small: 'size-12 rounded-2xl',
    medium: 'size-16 rounded-2xl',
    large: 'size-24 rounded-3xl'
  };

  return (
    <div className={sizeClasses[size]}>
      <img src={src} alt={alt} className="h-full w-full object-cover" />
    </div>
  );
};
```

Now reusable everywhere:

```
<Avatar src={user.avatar} alt={user.name} size="large" />
<Avatar src={commenter.avatar} alt={commenter.name} size="small" />
```

Step 6: Extract Repeated List Items

Create: `components/profile/ExperienceEntry.tsx`

```

import { MapPin, Ship, CalendarDays } from 'lucide-react';
import type { UserExperience } from '@types/user';

interface ExperienceEntryProps {
  experience: UserExperience;
}

export const ExperienceEntry = ({ experience }: ExperienceEntryProps) => {
  return (
    <article className="flex items-center gap-4 border-b border-gray-100 py-4">
      <div className="size-24 rounded-3xl bg-neutral-300" />
      <div className="flex flex-col gap-1.5">
        <h4 className="font-semibold">{experience.title}</h4>
        <p className="flex items-center gap-2 text-sm text-gray-600">
          <MapPin className="h-4 w-4" />
          {experience.location}
        </p>
        <p className="flex items-center gap-2 text-sm text-gray-600">
          <Ship className="h-4 w-4" />
          {experience.vessel}
        </p>
        <p className="flex items-center gap-2 text-sm text-gray-600">
          <CalendarDays className="h-4 w-4" />
          {format(new Date(experience.date), "do MMM yyyy")}
        </p>
      </div>
    </article>
  );
};

```

Step 7: Assemble with Semantic Structure

Final: `pages/Profile.tsx`


```

import { ProfileContainer } from '@components/layout/ProfileContainer';
import { ProfileHeader } from '@components/profile/ProfileHeader';
import { ProfileSection } from '@components/profile/ProfileSection';
import { ExperienceList } from '@components/profile/ExperienceList';
import { QualificationsList } from '@components/profile/QualificationsList';
import { FeedbackList } from '@components/profile/FeedbackList';
import { useUserProfile } from '@hooks/useUserProfile';

export const Profile = () => {
  const { user, loading, error } = useUserProfile();

  if (loading) return <LoadingSpinner />;
  if (error) return <ErrorMessage error={error} />;
  if (!user) return <NotFound />;

  return (
    <ProfileContainer>
      <ProfileHeader user={user} />

      <ProfileSection title="About & Hobbies">
        <p>{user.about}</p>
      </ProfileSection>

      <ProfileSection title="Experience Log">
        <ExperienceList experiences={user.experiences} />
      </ProfileSection>

      <ProfileSection title="Qualifications">
        <QualificationsList qualifications={user.qualifications} />
      </ProfileSection>

      <ProfileSection title="Skills">
        <p>{user.skills}</p>
      </ProfileSection>

      <ProfileSection title="Feedback">
        <FeedbackList feedback={user.feedback} />
      </ProfileSection>
    </ProfileContainer>
  );
};

```

Results:

- **60 lines** instead of 275 (78% reduction)
 - **Immediately understandable** structure
 - **Each component** < 100 lines
 - **Reusable** components
 - **Testable** in isolation
 - **Reviewable** by anyone
-

Advanced Pattern: Polymorphic Components

Allow components to render different HTML elements for semantic correctness:

```
interface CardProps {
  children: ReactNode;
  element?: 'div' | 'article' | 'section' | 'aside';
}

export const Card = ({ children, element: Element = 'div' }: CardProps) => {
  return (
    <Element className="rounded-lg border p-4 shadow-sm">
      {children}
    </Element>
  );
};

// Usage - semantically correct everywhere
<Card element="article">
  <BlogPost />
</Card>

<Card element="aside">
  <Sidebar />
</Card>
```

Recommended File Structure

```
src/
├── components/
│   ├── layout/                # Layout containers
│   │   ├── ProfileContainer.tsx
│   │   ├── PageContainer.tsx
│   │   └── ContentWrapper.tsx
│   ├── ui/                   # Reusable UI primitives
│   │   ├── Avatar.tsx
│   │   ├── Button.tsx
│   │   ├── StarRating.tsx
│   │   └── Card.tsx
│   └── profile/              # Domain-specific components
│       ├── ProfileHeader.tsx
│       ├── ProfileSection.tsx
│       ├── ExperienceEntry.tsx
│       └── ProfileActions.tsx
└── pages/
    └── Profile.tsx           # Page composition
```

Checklist: When to Extract a Component

Extract when:

- ☐ A pattern appears 3+ times
- ☐ A section has clear responsibility (header, card, list item)
- ☐ JSX block exceeds 20-30 lines
- ☐ You can name it in 1-2 words
- ☐ It could be reused elsewhere
- ☐ It helps with Single Level of Abstraction

Don't extract when:

- ☐ It's only used once and simple (< 10 lines)
 - ☐ The abstraction makes code harder to understand
 - ☐ It requires too many props (> 8)
-

Practice Exercise

Given this code, extract semantic components:

```
const JobPage = () => {
  return (
    <div className="px-24 py-6">
      <div className="flex flex-col gap-4">
        <div className="flex flex-row gap-2.5">
          <div className="relative size-24 rounded-3xl bg-neutral-300">
            <Heart className="absolute top-2.5 right-2.5 cursor-pointer" />
          </div>
          <div className="flex flex-col justify-between py-2">
            <h2 className="font-semibold">{job.title}</h2>
            <div className="flex items-center gap-x-4">
              <p className="flex items-center gap-1">
                <Clock className="h-3.5 w-3.5" /> {job.type}
              </p>
              <p className="flex items-center gap-1">
                <CalendarDays className="h-3.5 w-3.5" /> {job.date}
              </p>
            </div>
          </div>
        </div>
        <div className="ml-auto flex flex-col gap-4">
          <button className="rounded-xl bg-neutral-200">Share</button>
          <button className="rounded-xl bg-green-500">Apply</button>
        </div>
      </div>
    </div>
  );
};
```

Your task:

1. Identify semantic sections
2. Create component names
3. Extract components
4. Refactor the page

Target structure:

```
<JobPageContainer>
  <JobHeader job={job} />
</JobPageContainer>
```

Additional Resources

Articles

- [Semantic JSX by Max Böck](#)
- [React Component Design Principles](#)
- [Single Level of Abstraction Principle](#)

Best Practices

- Favor composition over configuration
 - Use semantic HTML elements
 - Keep components focused (Single Responsibility)
 - Name components by their purpose, not appearance
 - Make components testable and reusable
-

Key Takeaways

1. **Component names should communicate intent** - Not implementation details
2. **Structure over styling** - Code should be readable regardless of CSS approach
3. **One level of abstraction** - Don't mix high and low-level code
4. **DRY principle** - Extract repeated patterns
5. **Think in components** - Not in divs, styling, or CSS classes

Remember: The goal is readable, maintainable, reviewable code that anyone on the team can understand.

Something unclear in this section? Open an issue.